

An FPGA-Based Roadside Unit Controller for Deterministic Intersection Reservation in Autonomous Vehicle Networks

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Abstract—Coordinating vehicle access to shared intersections is a central problem in autonomous driving research, since uncoordinated arrivals from multiple directions can create unsafe conflicts. This paper presents the design, implementation, and verification of a Roadside Unit (RSU) intersection reservation controller realized in VHDL and deployed on a Xilinx Artix-7-based Nexys A7 FPGA board. The controller models a four-way intersection in which the North-South axis is designated the main road and the East-West axis the side road. Request signals from the four approaches are aggregated into two road groups and arbitrated by a three-state finite state machine (FSM) that enforces deterministic service windows, main-road priority, and mutually exclusive grant assignment between conflicting road groups. Functional correctness is established through a self-checking VHDL testbench executed in QuestaSim, and the design is further demonstrated on physical FPGA hardware using switches, LEDs, and seven-segment displays. The controller architecture is additionally extended toward physical realization through a hierarchical Altium Designer schematic and an accompanying three-dimensional printed circuit board (PCB) layout concept. Simulation results confirm that the controller satisfies the safety invariant that main-road and side-road grants are never asserted simultaneously, while priority and timing requirements are met as specified. The work establishes a complete and reproducible hardware design workflow, from requirements specification through simulation, FPGA prototyping, and PCB-level concept design, and identifies a concrete path toward a fully validated, communication-enabled RSU platform.

Index Terms—FPGA, VHDL, roadside unit, intersection reservation, finite state machine, autonomous vehicles, hardware verification, printed circuit board design, QuestaSim.

I. INTRODUCTION

Intersections are among the most safety-critical elements of road infrastructure, since they require multiple independent traffic streams to share a common spatial resource without collision. In conventional traffic systems, this coordination is achieved through fixed-time or actuated traffic signals interpreted by human drivers. In an autonomous driving context, however, coordination can instead be delegated to infrastructure-side computation: a Roadside Unit (RSU) can receive digital access requests from approaching vehicles and return deterministic grant decisions that authorize entry into the intersection's critical zone [5].

This paper reports the design and verification of a simplified, hardware-realized RSU controller targeting a four-way intersection. The controller accepts request signals from the North, South, East, and West approaches and produces

grant signals that authorize individual vehicles to proceed. The intersection center is modeled as a critical zone that must never be occupied by conflicting traffic streams simultaneously. The controller logic is described in VHDL, formally exercised through a dedicated testbench in QuestaSim, and subsequently deployed on a Nexys A7 development board, where switches emulate vehicle requests and LEDs together with seven-segment displays provide observable grant and status outputs. To demonstrate a complete hardware engineering workflow, the design is further carried into Altium Designer, where a hierarchical schematic and a three-dimensional PCB layout concept illustrate how the FPGA-based controller could be transferred to dedicated board hardware.

The remainder of this paper is organized as follows. Section II motivates the design choices. Section III describes the intersection scenario. Section IV states the project objectives. Section V summarizes the relevant technical background and related work. Sections VI–XI specify the system requirements, interface signals, timing parameters, safety invariant, and controller architecture. Section XII details the finite state machine. Sections XIII–XVII describe the VHDL implementation, FPGA prototype, simulation results, schematic design, and PCB layout. Section XVIII reports validation results, and Sections XIX–XXI provide discussion, outlook, and conclusions.

II. MOTIVATION

Uncoordinated access to an intersection by autonomous vehicles arriving from different directions creates a direct safety hazard, since two conflicting traffic streams occupying the critical zone concurrently would result in collision. Where a conventional traffic light communicates permission visually to a human operator, an autonomous-vehicle environment instead requires this permission to be issued and verified at the level of digital signals between infrastructure and vehicle.

Field-Programmable Gate Arrays (FPGAs) are well suited to this class of problem because the resulting control logic executes as dedicated hardware rather than as software running on a sequential processor, giving cycle-accurate, predictable timing behavior that can be exhaustively exercised in simulation prior to physical deployment [1], [2]. This motivates the present project: the implementation of a simplified but fully

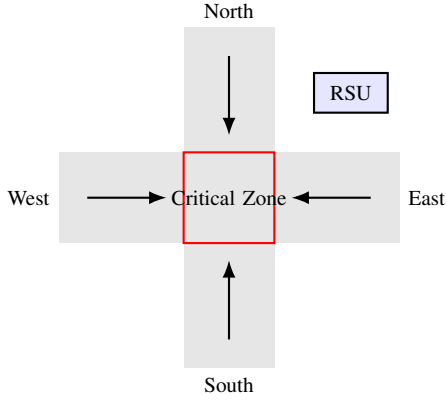


Fig. 1. Four-way intersection scenario with critical zone and RSU controller.

deterministic RSU controller in FPGA hardware, designed to satisfy three governing requirements:

- **Safety** – conflicting road groups must never receive a grant simultaneously.
- **Priority** – the main road must be served first whenever both road groups have active requests.
- **Determinism** – the timing behavior of the controller must be predictable, observable, and formally verifiable.

III. SYSTEM SCENARIO

The controller models a four-way intersection with incoming approaches from the North, South, East, and West, as illustrated in Fig. 1. The North-South axis is designated the main road, reflecting a higher-volume traffic corridor, while the East-West axis is designated the side road. The RSU, located at the intersection, receives request signals from all four approaches and arbitrates access to the shared critical zone at the intersection center.

Consistent with the project’s hardware focus, the controller does not compute detailed vehicle trajectories or continuous-time motion. Instead, it addresses the discrete decision problem of request handling and conflict-free grant generation, which constitutes the core arbitration logic required by any higher-level reservation scheme [5].

IV. PROJECT OBJECTIVES

The primary objective of this project is the design and hardware validation of an FPGA-based RSU controller for autonomous intersection reservation. The specific objectives are to:

- design a VHDL-based RSU controller that accepts vehicle request signals from four directions;
- group North and South requests as the main-road traffic stream;
- group East and West requests as the side-road traffic stream;
- grant priority to the main road whenever both road groups have active requests;
- serve the side road whenever no main-road request is pending;

- guarantee that conflicting road groups never receive simultaneous grants;
- verify the controller using QuestaSim waveform simulation;
- demonstrate the controller on a Nexys A7 FPGA board; and
- develop a schematic and PCB layout concept in Altium Designer.

V. TECHNICAL BACKGROUND AND RELATED WORK

A. FPGA and VHDL

An FPGA is a reconfigurable digital device that implements logic directly as hardware circuits rather than as instructions executed sequentially by a processor, which makes it suitable for control tasks that demand predictable, cycle-level timing behavior. VHDL is a hardware description language that allows the structure, interface, and timing behavior of such circuits to be specified, simulated, and subsequently synthesized for FPGA deployment [2], [3]. The combination of VHDL design entry with FPGA targets is a well-established methodology in digital systems engineering and underlies the controller developed in this work [1].

B. Finite State Machines for Digital Control

The finite state machine is a foundational structure for sequential digital control, in which a system occupies exactly one defined state at any time and transitions between states according to input conditions and internal timing [4]. FSMs are particularly well matched to traffic and intersection controllers, since the operating behavior of such systems naturally decomposes into a small number of clearly defined modes – in the present case, an idle mode together with main-road and side-road service modes.

C. Autonomous Intersection Management

Autonomous intersection management can be formulated as a request-and-permission problem, in which approaching vehicles request access to a shared intersection resource and a centralized or infrastructure-based controller determines whether each request can be granted safely. Dresner and Stone introduced a reservation-based framework for this problem, in which vehicles communicate with an intersection manager to negotiate space-time access to the intersection [5]. Communication between vehicles and roadside infrastructure in such systems is commonly envisioned over dedicated short-range or cellular vehicle-to-everything (V2X) links [6]. The controller developed in this paper follows the same conceptual request-and-grant pattern as these reservation-based approaches, but in a deliberately simplified, hardware-level form that omits explicit vehicle-to-infrastructure communication in favor of direct digital request inputs, leaving such communication as a defined extension for future work.

TABLE I
FUNCTIONAL REQUIREMENTS

ID	Requirement
FR1	The RSU shall receive request signals from North, South, East, and West.
FR2	The RSU shall group North and South as the main road.
FR3	The RSU shall group East and West as the side road.
FR4	The RSU shall grant the main road first when both main and side roads request access.
FR5	The RSU shall grant the side road when the main road has no active request.
FR6	The RSU shall prevent main-road and side-road grants at the same time.
FR7	The RSU shall indicate the active road group via a road-code output.
FR8	The RSU shall indicate the remaining service time via a countdown output.

TABLE II
NON-FUNCTIONAL REQUIREMENTS

ID	Requirement
NFR1	The design shall be implemented in VHDL.
NFR2	The design shall be suitable for FPGA implementation.
NFR3	The design shall be testable using a dedicated VHDL testbench.
NFR4	The design shall be observable through LEDs and seven-segment displays.
NFR5	The schematic and PCB concept shall support future hardware implementation.

TABLE III
INPUT SIGNALS

Signal	Type	Description
clk	Input	System clock signal.
reset	Input	Asynchronous reset; returns the controller to the IDLE state.
tick_1s	Input	One-second timing pulse driving the service timer.
N_req	Input	North request input.
S_req	Input	South request input.
E_req	Input	East request input.
W_req	Input	West request input.

VI. SYSTEM REQUIREMENTS

A. Functional Requirements

B. Non-Functional Requirements

VII. SYSTEM INTERFACE SIGNALS

A. Input Signals

The North and South requests are combined into a single main-road request, and the East and West requests are combined into a single side-road request, as expressed in (1) and (2):

$$\text{Main_req} = \text{N_req} \vee \text{S_req} \quad (1)$$

$$\text{Side_req} = \text{E_req} \vee \text{W_req} \quad (2)$$

TABLE IV
OUTPUT SIGNALS

Signal	Type	Description
N_grant	Output	Grant signal for North.
S_grant	Output	Grant signal for South.
E_grant	Output	Grant signal for East.
W_grant	Output	Grant signal for West.
road_code	Output	Indicates the currently active road group.
countdown	Output	Indicates the remaining service time.
main_req_wave	Output	Debug waveform for the main-road request.
side_req_wave	Output	Debug waveform for the side-road request.
main_grant_wave	Output	Debug waveform for the main-road grant.
side_grant_wave	Output	Debug waveform for the side-road grant.
conflict_wave	Output	Debug waveform flagging simultaneous main- and side-road requests.

TABLE V
TIMING PARAMETERS USED IN THE FINAL VHDL DESIGN

Parameter	Value	Meaning
MAIN_TIME	20	Main-road service window.
SIDE_TIME	10	Side-road service window.
tick_1s	1 pulse	Timer update signal.

B. Output Signals

Analogously, the individual grant outputs are combined into main-road and side-road grant signals, as expressed in (3) and (4):

$$\text{Main_grant} = \text{N_grant} \vee \text{S_grant} \quad (3)$$

$$\text{Side_grant} = \text{E_grant} \vee \text{W_grant} \quad (4)$$

VIII. TIMING REQUIREMENTS

The implemented controller uses fixed service-time counters to bound the duration of each service phase: the main road is served for 20 timer ticks, and the side road is served for 10 timer ticks, reflecting the higher expected traffic volume on the main road. During simulation, the `tick_1s` signal is generated synthetically at an accelerated rate to keep the resulting waveforms compact and readable. The timing parameters used in the final design are summarized in Table V.

IX. SAFETY INVARIANT

The controller's central safety requirement is that the main road and side road must never be granted access simultaneously, formalized as the invariant

$$(\text{N_grant} \vee \text{S_grant}) \wedge (\text{E_grant} \vee \text{W_grant}) = 0, \quad (5)$$

which is equivalent to the compact mutual-exclusion condition

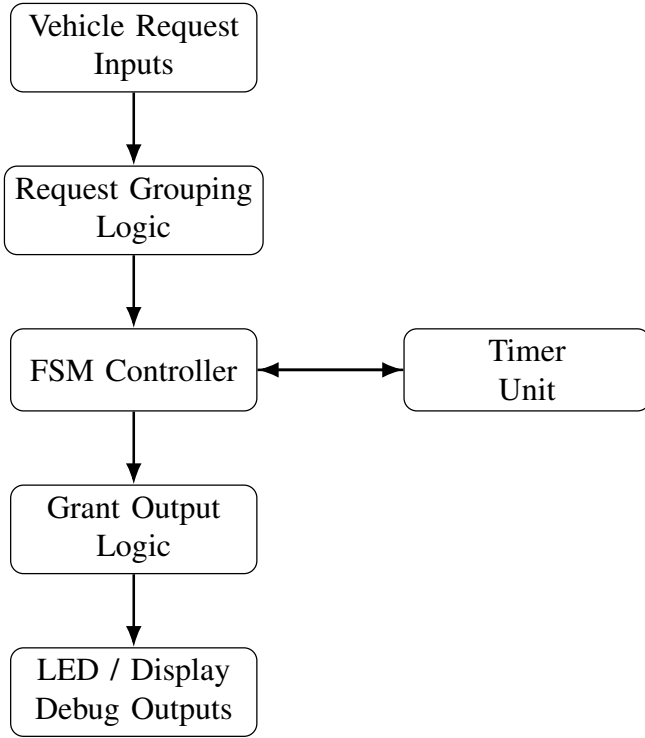


Fig. 2. Internal architecture of the FPGA-based RSU controller.

$$\text{Main_grant} \wedge \text{Side_grant} = 0. \quad (6)$$

In the implemented design, this invariant is enforced structurally rather than through runtime checking: North and South grant outputs can only be asserted while the FSM occupies the `MAIN_SERVICE` state, and East and West grant outputs can only be asserted while the FSM occupies the `SIDE_SERVICE` state. Because these two states are mutually exclusive by construction, the safety invariant holds by design and is additionally confirmed through simulation in Section XIV.

X. CONTROLLER ARCHITECTURE

The RSU controller is organized into five functional blocks: request grouping logic, a finite state machine, a timer unit, grant output logic, and debug/display outputs, as shown in Fig. 2. Request grouping logic combines the four directional requests into main-road and side-road signals, which drive the FSM. The FSM exchanges state and timing information with the timer unit to determine when a service window has expired, and its current state drives the grant output logic, which in turn produces both the directional grant signals and the LED/display debug outputs.

XI. FINITE STATE MACHINE DESIGN

The implemented controller uses a three-state FSM, consisting of the states `IDLE`, `MAIN_SERVICE`, and `SIDE_SERVICE`, summarized in Table VI and illustrated in Fig. 3. This three-state design was adopted in place of an earlier five-state concept after evaluation showed that

TABLE VI
FSM STATE DESCRIPTION

State	Description
<code>IDLE</code>	No active request is present.
<code>MAIN_SERVICE</code>	North/South main road is being served.
<code>SIDE_SERVICE</code>	East/West side road is being served.

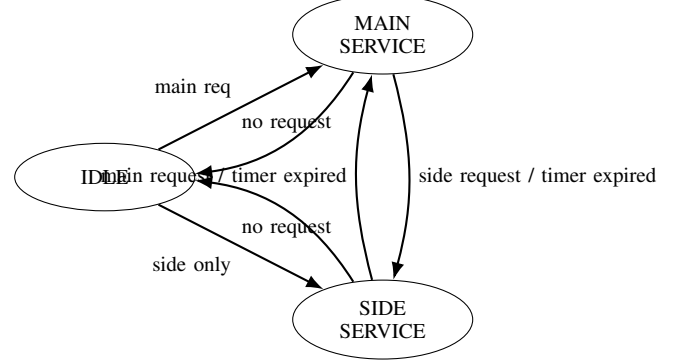


Fig. 3. Three-state FSM used in the final RSU controller.

the additional states introduced redundant transitions without improving safety or determinism.

A. State Transition Behavior

While in `IDLE`, the controller continuously evaluates the grouped request signals. A pending main-road request transitions the controller into `MAIN_SERVICE`; if only a side-road request is pending, the controller transitions into `SIDE_SERVICE`. During `MAIN_SERVICE`, the North and South grant outputs are asserted according to the individual `N_req` and `S_req` inputs, and the controller transitions to `SIDE_SERVICE` once the main-road timer expires and a side-road request is pending, or returns to `IDLE` if no further requests remain. The symmetric behavior applies to `SIDE_SERVICE` with respect to the East and West grant outputs.

XII. VHDL IMPLEMENTATION

The controller is implemented as a single VHDL entity with clock, reset, request, and grant ports, together with debug waveform outputs, a road-code output, and a countdown output. The complete VHDL source code and accompanying testbench are submitted separately with the project files and are omitted from this report to keep the discussion focused on design rationale rather than implementation listing.

The central design decision is the reduction of four independent directional requests to two arbitrated road-group requests, as defined in (1) and (2). The FSM state then exclusively determines which road group is permitted to assert grant outputs, which directly enforces the safety invariant of (5)–(6). The `road_code` output exposes the active FSM state for external observation, with the encoding summarized in Table VII.

TABLE VII
ROAD CODE ENCODING

Code	Meaning
00	Idle state.
01	Main-road service.
10	Side-road service.

TABLE VIII
FPGA PROTOTYPE INPUT/OUTPUT MAPPING

FPGA Element	Signal	Purpose
SW0	N_req	North request.
SW1	S_req	South request.
SW2	E_req	East request.
SW3	W_req	West request.
LED1	N_grant	North grant.
LED3	S_grant	South grant.
LED2	E_grant	East grant.
LED4	W_grant	West grant.
Seven-segment	Debug	Road code and countdown display.

TABLE IX
SIMULATION TEST CASES

TC	Input Condition	Expected Result
TC1	No request	No grant active.
TC2	North/South request	Main-road grant active.
TC3	East/West request	Side-road grant active.
TC4	Main and side request together	Main road served first.
TC5	Timer expires while both groups request	Controller switches active road group.
TC6	Main and side grants checked jointly	Conflicting grants never occur together.

XIII. FPGA PROTOTYPE IMPLEMENTATION

The controller was deployed on a Nexys A7 development board, which is built around a Xilinx Artix-7 FPGA and provides on-board switches, LEDs, and seven-segment displays suitable for rapid educational prototyping [7], [8]. Board switches were mapped to the four directional request inputs, LEDs were mapped to the corresponding grant outputs, and the seven-segment display was used to present the road code and countdown values for visual debugging. The complete input/output mapping is given in Table VIII.

XIV. SIMULATION AND VERIFICATION

Functional correctness of the controller was verified using a dedicated VHDL testbench executed in QuestaSim. The testbench applies a sequence of request scenarios and records the resulting grant outputs, FSM state, road code, and countdown timer for inspection.

A. Verification Test Cases

Six representative test cases were defined to exercise the controller's core functional and safety requirements, summarized in Table IX.



Fig. 4. Nexys A7 FPGA prototype used to demonstrate the RSU controller via switches, LEDs, and seven-segment display.

Fig. 5 shows the request input signals together with the corresponding grant output signals across the simulated scenarios, indexed by the *scenario_id* signal. The grant outputs are observed to track the active requests consistently with the controller's defined state behavior.

Fig. 6 illustrates the internal controller behavior, where the *road_code* output encodes the active road group (00: idle, 01: main-road service, 10: side-road service) and the countdown signal tracks the remaining service time for the currently active road group. Across all six test cases, the main-road and side-road grant signals were never observed to be active simultaneously, confirming the safety invariant of (5)–(6).

XV. SCHEMATIC DESIGN

To extend the controller toward a dedicated hardware platform, a hierarchical schematic was developed in Altium Designer, comprising power, FPGA core, input/output, and debug display sheets. Hierarchical decomposition was adopted because it partitions the overall system into manageable, independently reviewable functional blocks [9].

A. Top-Level Schematic

The top-level schematic connects the major functional sheets through hierarchical sheet symbols, providing a single-page overview of the complete board architecture.

B. Power Schematic

The power schematic provides the regulated supply rails required by the FPGA-based board, including input power

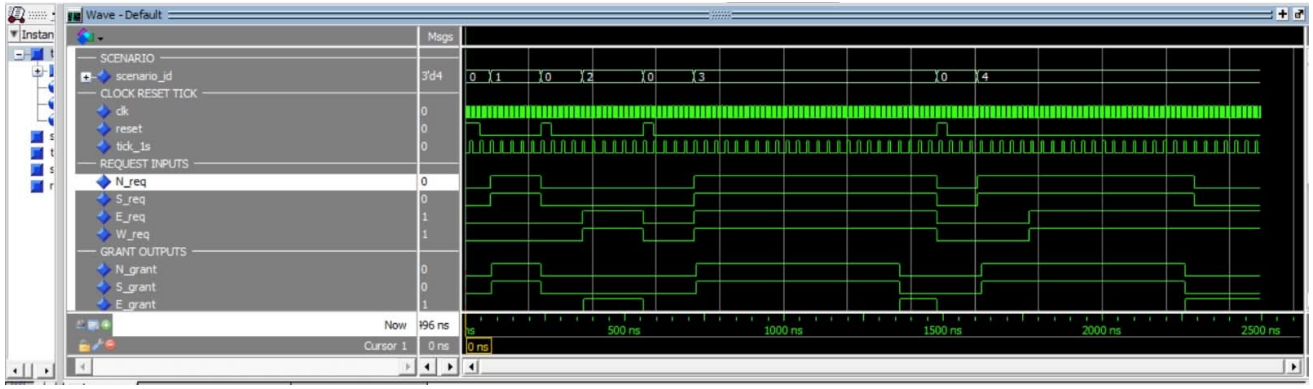


Fig. 5. QuestaSim waveform showing request inputs and grant outputs for different traffic scenarios.

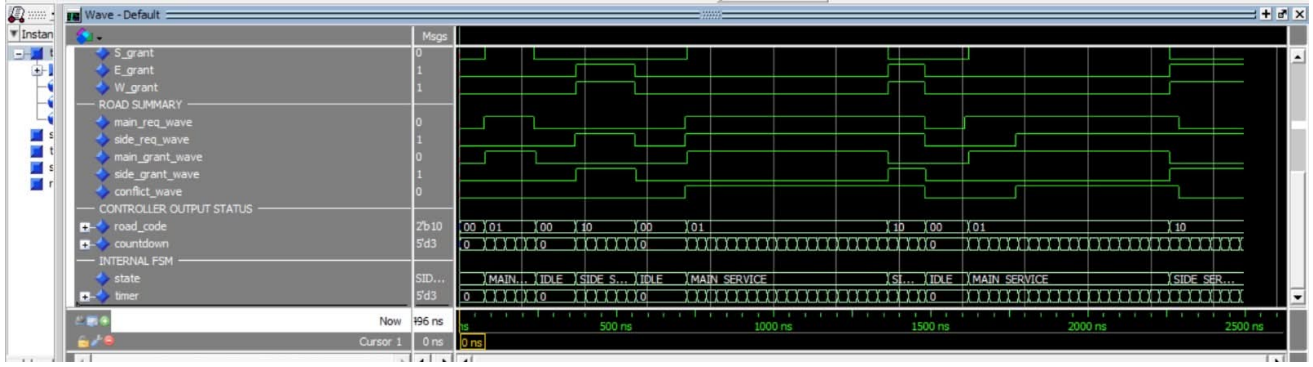


Fig. 6. QuestaSim waveform showing road summary signals, road code, countdown timer, and internal FSM state.

protection, voltage regulator circuits, a power-indicator LED, and dedicated test points for bring-up and debugging.

C. FPGA Core Schematic

The FPGA core schematic contains the FPGA device together with its required support circuitry, including the clock source, reset network, JTAG programming connector, power pins, and primary I/O signal connections.

D. Input and Output Schematic

The input/output schematic connects the four directional request switches and the four corresponding grant LEDs. Pull-down resistors establish well-defined logic levels at the request inputs, and current-limiting resistors protect the grant LEDs.

E. Debug Display Schematic

The debug display schematic provides additional visual feedback through seven-segment displays and integrates the reset circuit, clock oscillator, and JTAG-related connections required for programming and debugging.

XVI. PCB LAYOUT DESIGN

A PCB layout concept was developed to illustrate the physical organization of the RSU controller board, with direction labels (North, South, East, West) used to make the board's correspondence to the intersection model explicit. PCB design

constitutes the essential bridge between an electrically correct schematic and a manufacturable physical artifact [10].

Fig. 12 shows the resulting physical layout. Request switches and grant LEDs are positioned adjacent to their corresponding directional labels, while the FPGA device, power input connector, programming connector, and debug displays are placed in accessible locations to facilitate testing and demonstration.

XVII. BILL OF MATERIALS

XVIII. RESULTS

A. Simulation Results

QuestaSim simulation confirmed that the controller follows the specified state transitions and timing behavior across all defined test cases. The grant outputs were observed to activate correctly with respect to the prevailing request inputs and active FSM state, and the road code and countdown outputs consistently reflected the controller's internal timing.

B. FPGA Prototype Results

The Nexys A7 FPGA prototype reproduced the controller's specified behavior using physical switches, LEDs, and seven-segment displays. Manually applied request inputs produced the expected grant outputs, providing an independent hardware-level confirmation of the simulation results.

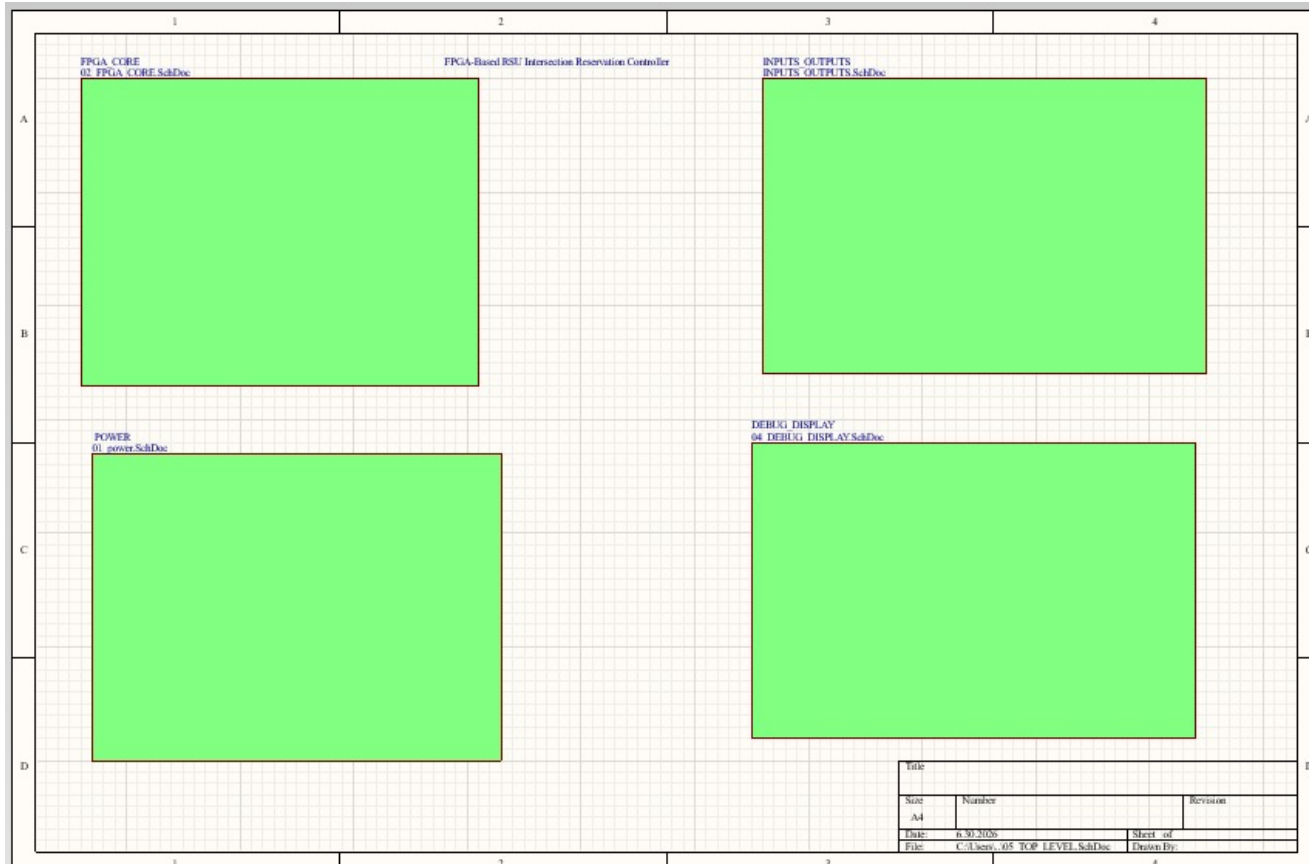


Fig. 7. Top-level hierarchical schematic showing FPGA core, input/output, power, and debug display sheets.

TABLE X
MAIN BILL OF MATERIALS

No.	Component	Qty.	Purpose	Requirement	Status	Evidence
1	FPGA device	1	Main controller.	Main-road priority	Passed	QuestaSim waveform.
2	Clock oscillator	1	System clock.	Side-road service	Passed	FSM and countdown wave-
3	Reset button	1	Reset input.			form.
4	Switches/buttons	4	Vehicle request in-	No conflicting grants	Passed	State-based grant logic.
5	LEDs	4	puts.	FPGA implementation	Passed	Nexys A7 demonstration.
6	Seven-segment displays	2	Grant outputs.	Schematic design	Completed	Altium schematic sheets.
7	Resistors	Several	Debug display.	PCB concept	Completed	3D PCB view.
8	Capacitors	Several	Pull-down and LED current limiting.			
9	Voltage regulators	Several	Power filtering and decoupling.			
10	JTAG connector	1	Power supply rails.			
			FPGA programming/debug.			

TABLE XI
REQUIREMENT VALIDATION SUMMARY

D. Requirement Validation
Table XI summarizes the validation status of the principal system requirements against the available evidence.

XIX. DISCUSSION

The implemented RSU controller demonstrates that the core arbitration logic of an autonomous intersection reservation scheme can be realized as deterministic digital hardware. The FSM-based architecture yields a design that is readable, exhaustively testable, and directly synthesizable for FPGA deployment, and the chosen safety strategy – permitting only one road group to be active at any time – enforces the required mutual-exclusion property by construction rather than by post-hoc checking.

C. Schematic and PCB Results

The Altium schematic was organized into separate, clearly defined hierarchical sheets, and the resulting 3D PCB view demonstrates that the design can be physically arranged as a recognizable intersection-model board, with direction labels and clearly visible input/output components.

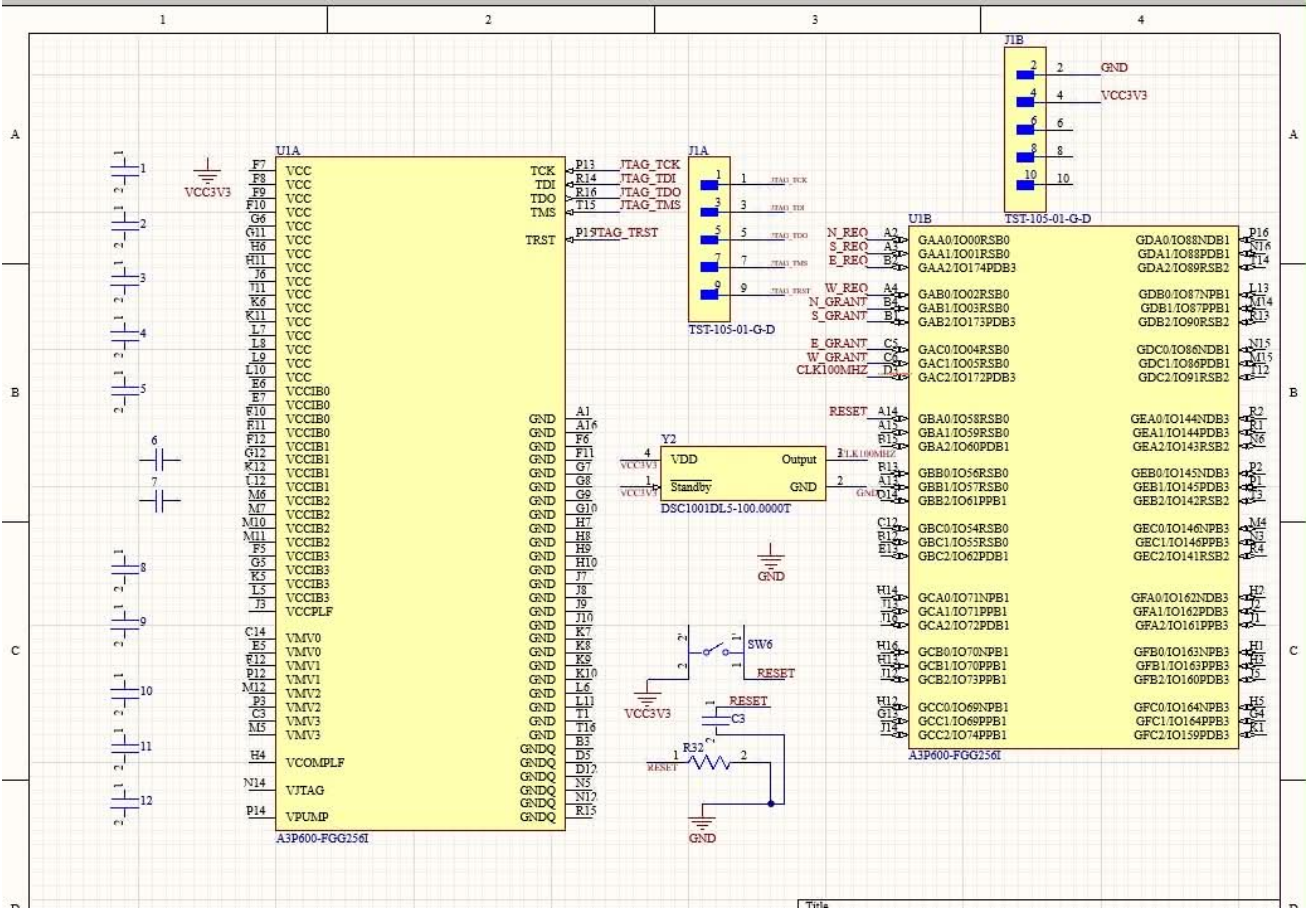


Fig. 9. FPGA core schematic containing the FPGA device, clock source, reset circuit, JTAG connector, and signal connections.

A further significant extension is the introduction of wireless V2X communication [6], in which vehicles transmit digital request messages directly to the RSU rather than relying on discrete switch inputs; this would allow grant decisions to be computed from vehicle position, speed, heading, priority class, and estimated time of arrival, consistent with reservation-based intersection management frameworks in the literature [5]. On the hardware side, the PCB concept developed in this work could be carried forward toward manufacturing through electrical rule checking, design rule checking, footprint verification, Gerber file generation, bill-of-materials validation, assembly, and full hardware testing.

In summary, future work may include:

- adding left-turn, right-turn, and straight movement requests;
- adding emergency-vehicle priority;
- adding wireless V2X communication;
- adding vehicle speed and arrival-time estimation;
- adding fairness and starvation-prevention logic;
- adding formal verification of the safety property;
- manufacturing and electrically testing the custom PCB; and
- improving the seven-segment display interface.

XXII. CONCLUSION

This paper presented the design, implementation, and verification of an FPGA-based RSU intersection reservation controller for autonomous vehicles. The controller accepts request signals from the North, South, East, and West directions and produces grant signals according to a defined priority, timing, and safety scheme, with the North-South road treated as the main road and the East-West road treated as the side road.

The VHDL implementation was verified through a dedicated testbench in QuestaSim, and the resulting waveforms confirmed that the controller follows the expected state transitions while never asserting conflicting main-road and side-road grants simultaneously. The design was further demonstrated on a Nexys A7 FPGA board, where switches, LEDs, and seven-segment displays were used to observe controller behavior directly on hardware.

The accompanying Altium schematic and PCB layout concept illustrate a credible path toward a dedicated hardware board for the controller. Although the PCB has not yet been manufactured, the schematic sheets and three-dimensional PCB view provide a clear and well-organized hardware design direction. Overall, this work demonstrates a complete hardware engineering design workflow – spanning requirements

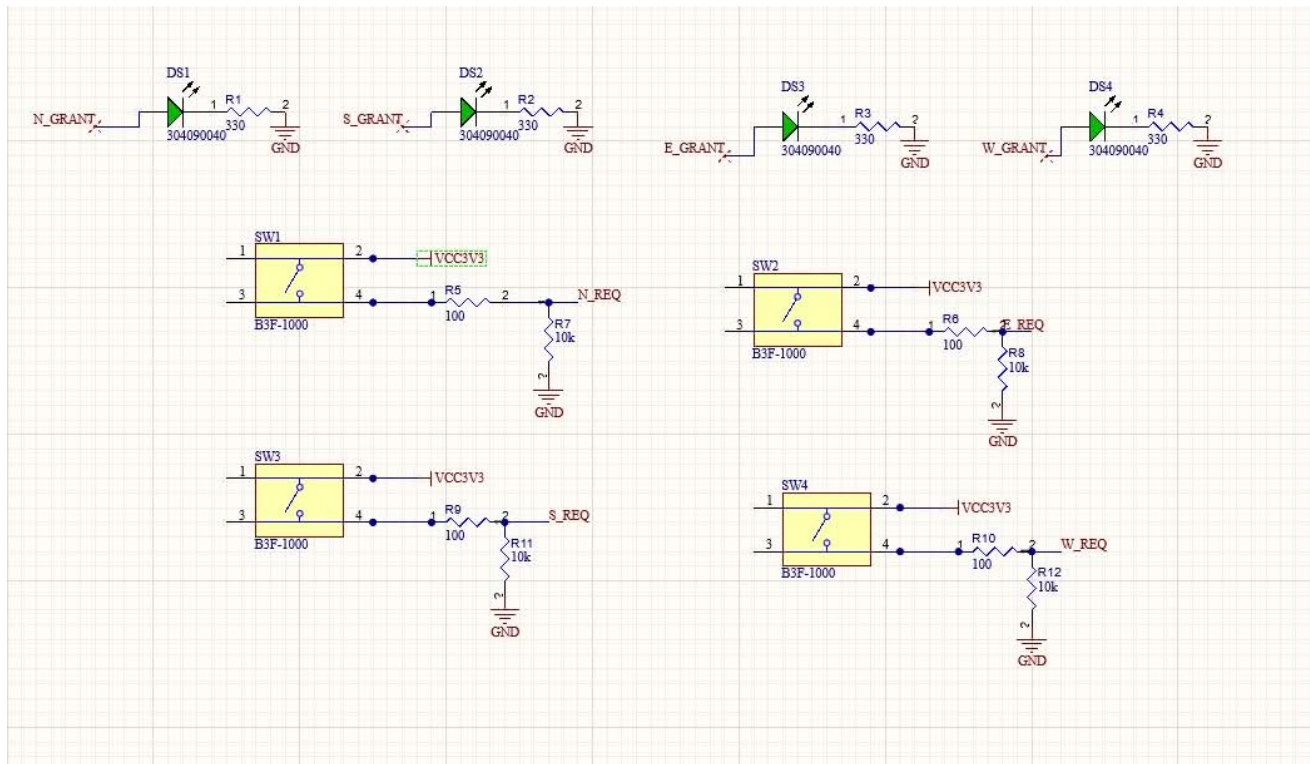


Fig. 10. Input and output schematic showing request switches and grant LEDs for North, South, East, and West directions.

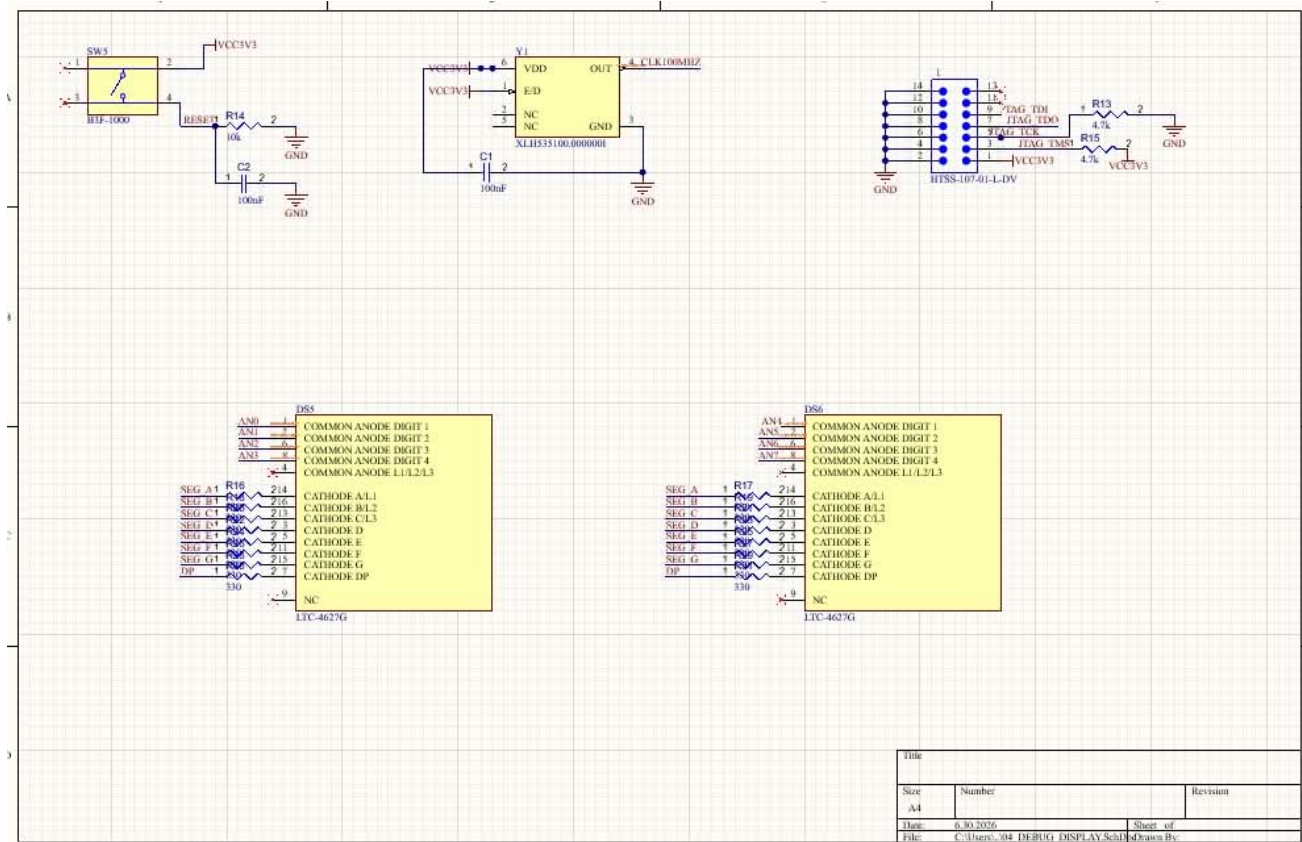


Fig. 11. Debug display schematic with seven-segment displays, reset input, clock oscillator, and JTAG interface.

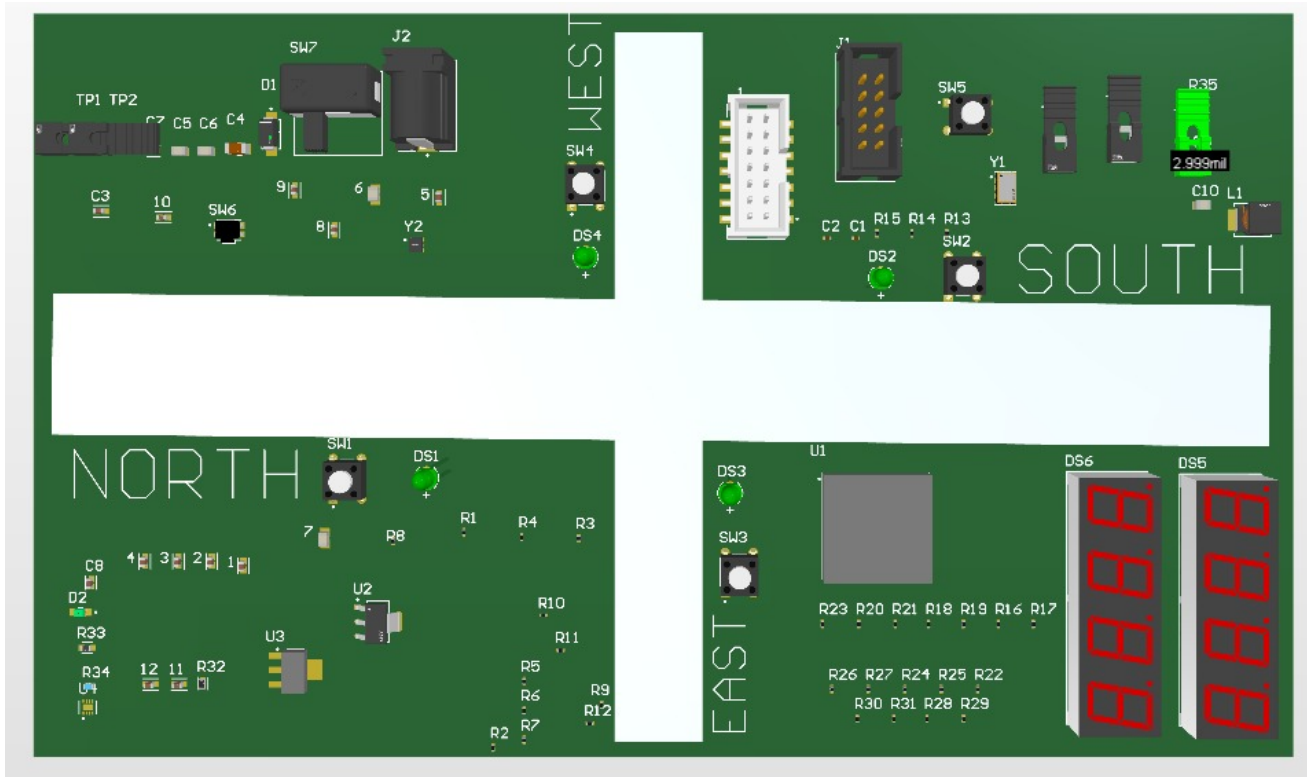


Fig. 12. 3D PCB view of the FPGA-based RSU controller board with direction labels, switches, LEDs, power input, JTAG connector, FPGA, and seven-segment displays.

analysis, VHDL-based controller design, simulation, FPGA prototype testing, schematic design, and PCB layout planning – and provides a sound foundation for a more advanced RSU system incorporating V2X communication, emergency priority, and manufactured PCB hardware.

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APPENDIX A SUBMITTED PROJECT FILES

The complete source files are omitted from this report to preserve readability and are submitted separately as part of the project package:

- `rsu_controller.vhd`
- `tb_rsu_controller.vhd`
- Vivado project files
- QuestaSim simulation files
- Altium schematic files
- PCB layout files

APPENDIX B REPORT FIGURE FILES

The following image files are referenced in this report and must be placed inside a `figures` subfolder relative to the `.tex` source file:

- `fig_power_schematic.jpg`
- `fig_questasim_request_grant.jpg`
- `fig_nexys_a7_demo.jpg`
- `fig_questasim_fsm_status.jpg`
- `fig_io_schematic.jpg`
- `fig_fpga_core_schematic.jpg`
- `fig_top_level_schematic.jpg`
- `fig_debug_display_schematic.jpg`
- `fig_pcb_3d_view.jpg`